

**SIMATS ENGINEERING**

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**

**CHENNAI – 602105**

**Department of Computer Science and Engineering**

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**ASSESSMENT TOOL 3 – Visualization Design Activity**

|                     |         |                      |                                              |
|---------------------|---------|----------------------|----------------------------------------------|
| <b>Course Code:</b> | DSA0502 | <b>Course Title:</b> | Query Processing for Data Science            |
| <b>CO Assessed:</b> | CO5     | <b>Assessment:</b>   | Tool 3– <b>Visualization Design Activity</b> |
| <b>Weightage:</b>   | 35%     | <b>Total Marks:</b>  |                                              |

**CO5 Statement:** Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

|                           |                             |                            |                       |
|---------------------------|-----------------------------|----------------------------|-----------------------|
| <b>Student Name:</b>      | Sandra Sudevan              | <b>Reg. No.:</b>           | 192424422             |
| <b>Section Batch:</b> /   | 2014                        | <b>Date of Submission:</b> | 20/08/2026            |
| <b>Faculty In-charge:</b> | <b>Dr. B.Shyamala Gowri</b> | <b>Project Mode:</b>       | Individual Submission |

**Code of Conduct**

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandrasudevan

| No. | Activity                             | Task                                                                                                                                                 | Student Output                 | Time   |
|-----|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------|
| 1   | <b>Basic Plot Creation</b>           | Create a line plot using a small dataset of monthly sales. Add a title, x-axis label, y-axis label, legend, and grid.                                | Annotated line plot            | 15 min |
| 2   | <b>Plot Component Challenge</b>      | Given an incomplete Matplotlib program, identify and add the missing title, labels, legend, markers, and grid.                                       | Corrected Python program       | 10 min |
| 3   | <b>Compare Visualization Types</b>   | Use the same dataset to create a <b>line plot, bar plot, and scatter plot</b> . Discuss which visualization communicates the data most effectively.  | 3 plots + justification        | 15 min |
| 4   | <b>Pandas Visualization Activity</b> | Create a DataFrame containing Product, Sales, and Profit. Generate appropriate visualizations directly using Pandas.                                 | DataFrame + plots              | 15 min |
| 5   | <b>Trend Visualization</b>           | Use monthly/weekly data to identify increasing, decreasing, and fluctuating trends using a line plot.                                                | Trend visualization + analysis | 15 min |
| 6   | <b>Relationship Visualization</b>    | Plot two variables such as <b>Advertising vs Sales</b> using a scatter plot. Determine whether the relationship appears positive, negative, or weak. | Scatter plot + interpretation  | 15 min |
| 7   | <b>Distribution Visualization</b>    | Create a histogram of student marks or employee salaries. Identify the approximate shape, center, spread, and possible skewness.                     | Histogram + observations       | 15 min |
| 8   | <b>Frequency Visualization</b>       | Create a categorical dataset such as department-wise employee count. Represent the                                                                   | Frequency chart                | 10 min |

| No. | Activity                      | Task                                                                                                                                    | Student Output                          | Time   |
|-----|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------|
|     |                               | frequencies using a bar chart.                                                                                                          |                                         |        |
| 9   | Scatter Matrix Visualization  | Use a dataset containing at least four numerical variables. Generate a scatter matrix and identify the strongest visible relationships. | Scatter matrix + findings               | 20 min |
| 10  | Lag Plot Visualization        | Use time-series data to create a lag plot. Examine whether the current observation appears related to the previous observation.         | Lag plot + conclusion                   | 15 min |
| 11  | Autocorrelation Visualization | Generate an autocorrelation plot for a time series. Identify lags with relatively strong positive or negative autocorrelation.          | Autocorrelation plot + analysis         | 20 min |
| 12  | Bootstrap Visualization       | Select a numerical dataset, repeatedly resample it, and examine the distribution of a chosen statistic such as the mean.                | Bootstrap distribution + interpretation | 25 min |

### 3. Compare Visualization Types

#### Code:

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
month = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
```

```
sales = [120, 150, 140, 180, 210, 190]
```

```
# Line Plot
```

```
plt.figure()
```

```
plt.plot(month, sales, marker='o')
```

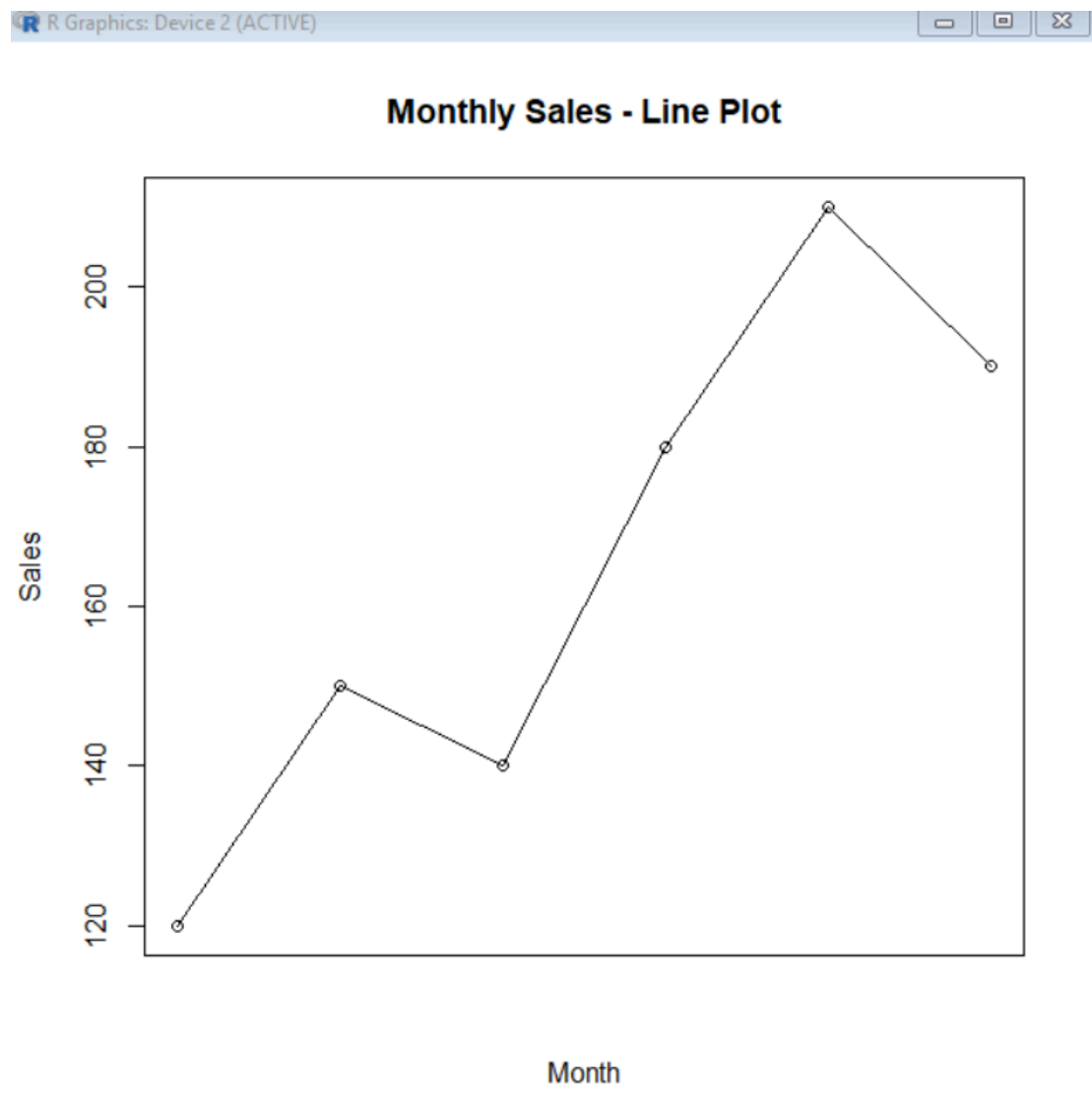
```
plt.title("Monthly Sales - Line Plot")
```

```
plt.xlabel("Month")
```

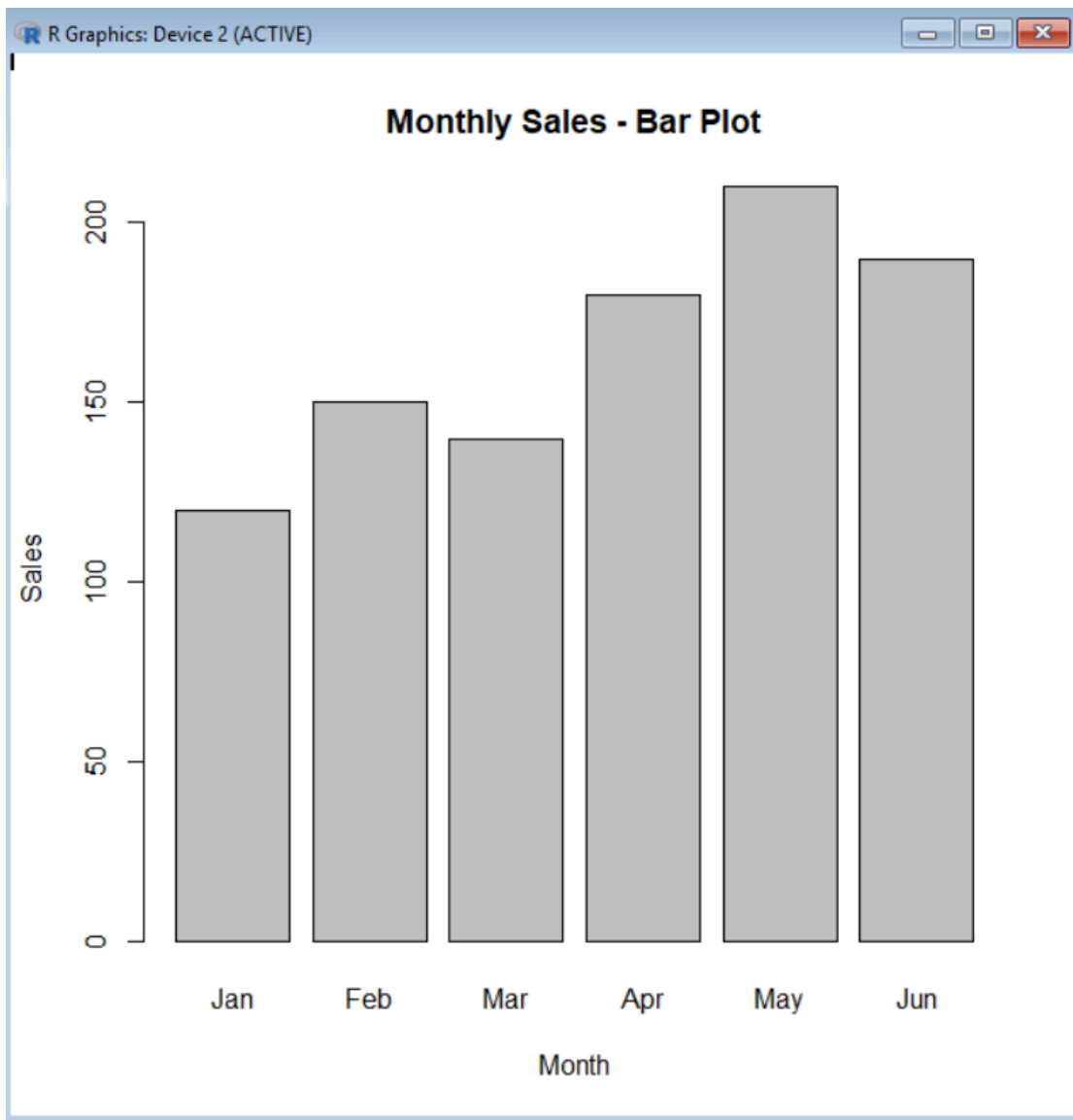
```
plt.ylabel("Sales")  
plt.show()
```

```
# Bar Plot  
plt.figure()  
plt.bar(month, sales)  
plt.title("Monthly Sales - Bar Plot")  
plt.xlabel("Month")  
plt.ylabel("Sales")  
plt.show()
```

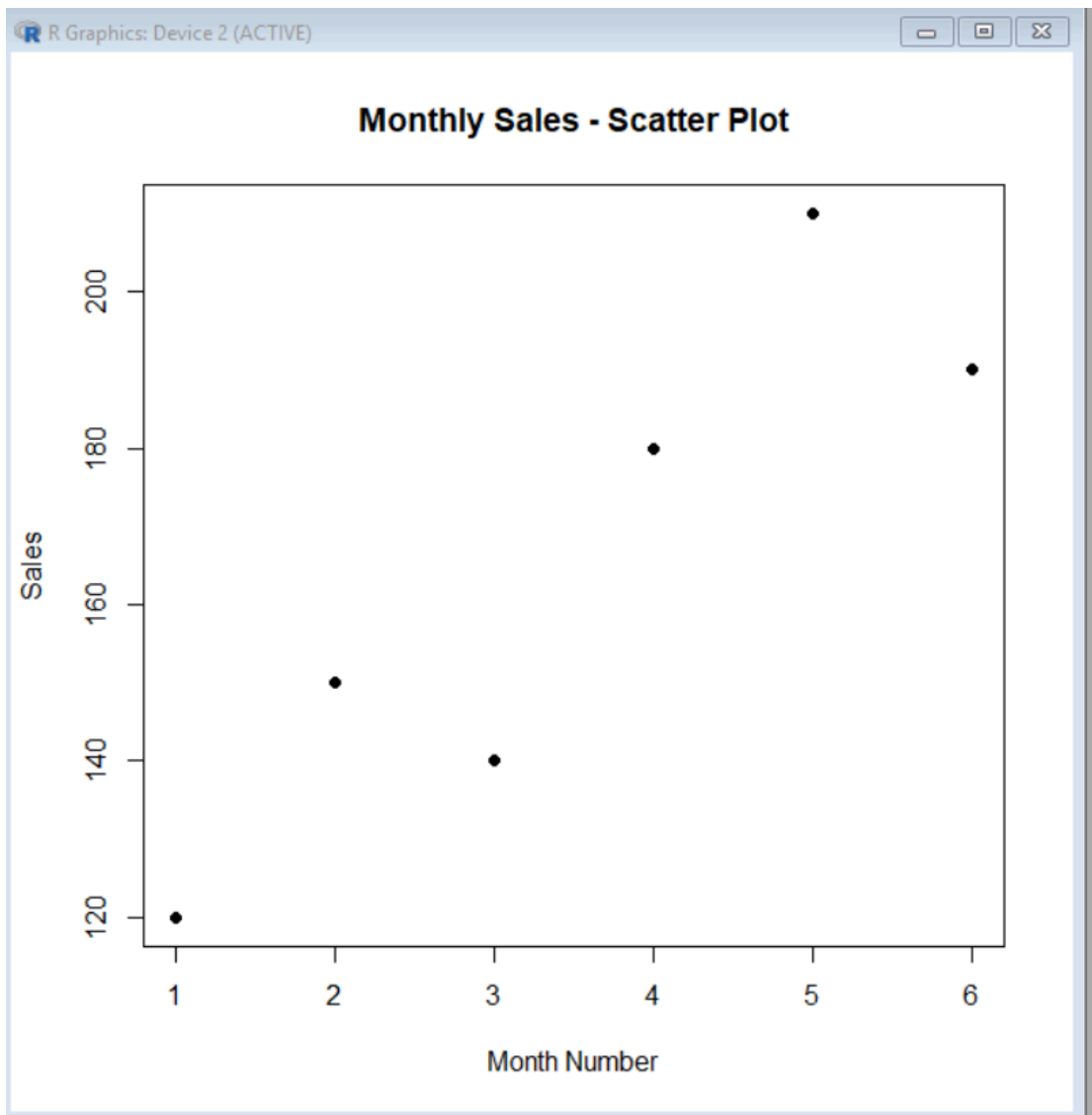
```
# Scatter Plot  
plt.figure()  
plt.scatter(month, sales)  
plt.title("Monthly Sales - Scatter Plot")  
plt.xlabel("Month")  
plt.ylabel("Sales")  
plt.show()
```



**Observation:** Clearly shows the trend and changes over time.



**Observation:** Makes it easy to compare individual months and identify the highest sales.



A scatter plot shows the sales values as separate points. It is useful for seeing the values, but it is not very easy to understand the increase or decrease from one month to another.

## 1. Basic Plot Creation

Code:

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
month = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
```

```
sales = [120, 150, 140, 180, 210, 190]
```

```
# Line Plot
```

```
plt.plot(month, sales, marker='o', label="Sales")
```

```
# Title and labels
```

```
plt.title("Monthly Sales")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales")
```

```
# Grid  
plt.grid(True)
```

```
# Legend  
plt.legend()
```

```
# Show plot  
plt.show()
```

**Title:** Monthly Sales

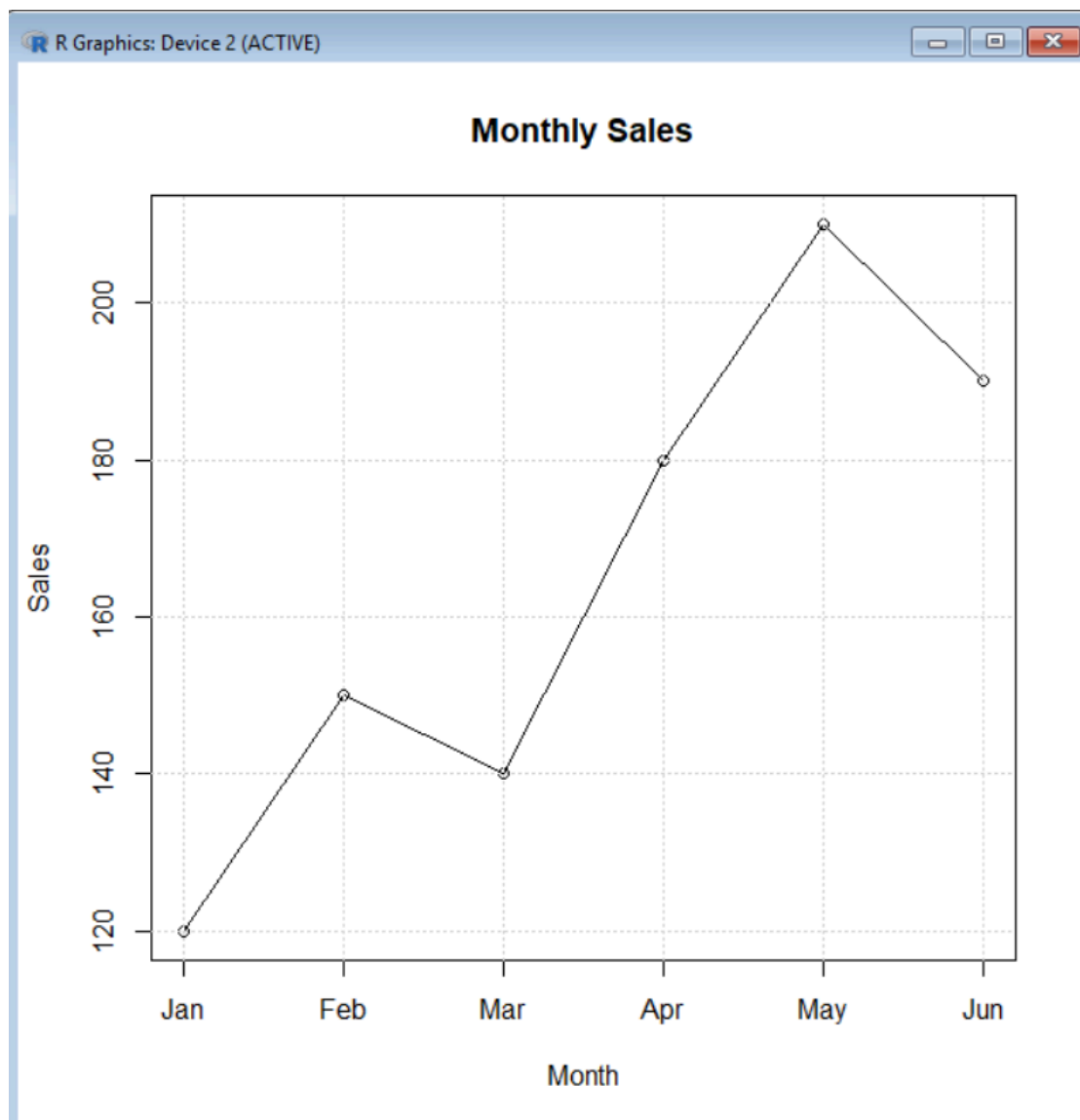
**X-axis label:** Month

**Y-axis label:** Sales

**Legend:** Sales

**Grid:** Makes the values easier to read

**Line + points:** Shows how sales change each month





The graph shows that sales increase from January to May. Sales reach the highest value of 210 in May and then decrease slightly to 190 in June.